Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<tool_call> reasoning process here

To find the de Broglie wavelength of an electron given its kinetic energy, we use the de Broglie wavelength formula:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ10âģ»Â³âģ´ JÂ·s (or 6.626 ÃĹ10âģ»Â²âģ· ergÂ·s),
- p = momentum of the electron.

For a non-relativistic electron, kinetic energy (K.E.) = (1/2)mvÂ², and the momentum p = mv.

So, p = âĪļ(2mK.E.)

Therefore, de Broglie wavelength:

Î» = h / âĪļ(2mK.E.)

We need to compute this with K.E. = 1 eV = 1.602 ÃĹ10âģ»Â¹âģ¹ J.

Mass of electron, m = 9.109 ÃĹ10âģ»Â³Â¹ kg

Step-by-step:

1. Compute 2mK.E.:

2 ÃĹ (9.109 ÃĹ10âģ»Â³Â¹) ÃĹ (1.602 ÃĹ10âģ»Â¹âģ¹) = 2.917 ÃĹ10âģ»âģµâģ° kgÂ·J

2. âĪļ(2mK.E.) = âĪļ(2.917 ÃĹ10âģ»âģµâģ°) âīĪ 1.708 ÃĹ10âģ»Â²âģµ kgÂ·m/s

3. h = 6.626 ÃĹ10âģ»Â³âģ´ JÂ·s

4. Î» = 6.626 ÃĹ10âģ»Â³âģ´ / 1.708 ÃĹ10âģ»Â²âģµ âīĪ 3.88 ÃĹ10âģ»âģ¹ m

Convert to cm: 3.88 ÃĹ10âģ»âģ¹ m = 3.88 ÃĹ10âģ»âģ· cm

Wait âĢĶ let's recheck units.

Actually, 1 m = 100 cm âĨĴ so 3.88 ÃĹ10âģ»âģ¹ m = 3.88 ÃĹ10âģ»âģ· cm

But look at the options: the choices are in cm, but one is given in m.

LetâĢĻs convert to cm:

3.88 ÃĹ10âģ»âģ¹ m = 3.88 ÃĹ10âģ»âģ· cm

Wait âĢĶ that's not matching any of the options.

Wait âĢĶ 3.88 ÃĹ10âģ»âģ¹ m is 3.88 ÃĹ10âģ»âģ· cm? Yes.

Check if option F is in meters: F. 5.27 ÃĹ10âģ»âģ¹ m âĨĴ is that close?

Wait, letâĢĻs double-check the calculation.

Use standard formula for electron wavelength in meters:

Î» (in m) = âĪļ(2 ÃĹ 9.11 ÃĹ10âģ»Â³Â¹ ÃĹ 1.602 ÃĹ10âģ»Â¹âģ¹) / (6.626 ÃĹ10âģ»Â³âģ´)

Wait better use standard shortcut:

Î» = h / âĪļ(2mE)

Compute numerically:

m = 9.11e-31 kg
E = 1.602e-19 J

So p = âĪļ(2 ÃĹ 9.11e-31 ÃĹ 1.602e-19) = âĪļ(2.917e-49) = 5.401e-25 kgÂ·m/s

h = 6.626e-34

Î» = 6.626e-34 / 5.401e-25 = 1.227e-9 m